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قســـــم هندسة البرمجيات ونظم المعلومات

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Academic Community Platform

(NexaCampus AI)

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Abstract

The rapid growth of digital technologies has significantly transformed the educational environment, creating a demand for secure and intelligent academic platforms that facilitate communication, collaboration, and knowledge sharing among university communities. Traditional communication methods often lack integration, security, and dedicated academic features, making it difficult for students and faculty members to interact efficiently within a unified environment.

This project presents **NexaCampus AI** , a secure academic community platform specifically designed to address the fragmentation of university learning resources, academic communication, and student services across separate tools. The proposed system combines faculty-specific academic portals, a learning management environment, social networking, research collaboration, and AI-supported learning assistance. Its intended users are students, professors, faculty administrators, and system administrators, each operating under role-specific permissions. The platform supports course administration, enrollment, lectures, assignments, examinations, resources, discussions, events, notifications, academic projects, and administration workflows. Nexa AI is intended to support course-aware question answering using retrieval-augmented generation (RAG), practice-quiz generation, and personalized study support.

The system was developed using the MERN Stack, consisting of MongoDB, Express.js, React.js, and Node.js, following a modern client-server architecture. Redux Toolkit was used for state management, while Clerk handled authentication and session management. The platform also incorporates responsive user interface design, secure RESTful APIs, real-time communication, and role-based access control to provide a reliable and scalable solution.
The implementation demonstrates that NexaCampus AI successfully satisfies the functional and non-functional requirements of an academic social networking platform. The developed system offers a secure authentication mechanism, an intuitive user experience, efficient real-time communication, and scalable architecture suitable for future enhancements. The project contributes a practical solution that strengthens academic collaboration while maintaining security, usability, and maintainability.

This report presents the problem, motivation, existing approaches, research gap, development methodology, requirements, feasibility, risks, and a proposed delivery plan.

Keywords: Learning Management System; Academic Social Network; Role-Based Access Control; Retrieval-Augmented Generation.

شــكر و تقديــر

اكتب فقرة موجزة لشكر المشرف، أعضاء الهيئة التدريسية، الجهة الداعمة، أو كل من أسهم في إنجاز المشروع.

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الاختصار	باللغة الإنجليزية	المعنى بالعربية

Chapter One – Introduction

Chapter One – Introduction

1.1 Project BackGround

Universities increasingly use digital platforms to distribute lectures, publish assignments, conduct assessments, and communicate with students. These functions may be available across learning management systems, collaboration tools, messaging applications, and departmental channels. When they are not coordinated in one experience, students must move between services to find course material, discuss an assignment, check an announcement, or contact a professor. The associated problem is not necessarily the absence of technology, but the lack of consistent academic context and predictable access to related information.

NexaCampus AI is proposed as an intelligent academic community platform for the Syrian Private University context. Its design brings together a learning management system (LMS), a university-oriented social network, faculty-specific services, research collaboration, and AI-supported learning assistance. The system is organized around four main user roles: Student, Professor, Faculty Admin, and System Admin. Each role is connected to a distinct set of academic and administrative responsibilities rather than relying on a single unrestricted account type.

1.2 Definition of the Project Problem

The central problem is the fragmentation of academic activities and communication into disconnected workflows. A student may receive a course announcement in one location, retrieve a PDF in another, submit coursework through a separate interface, and request clarification through informal messages. Professors face similar fragmentation when coordinating enrollment, course materials, assessments, and announcements, while administrators require consistent identity verification and oversight across faculties.

NexaCampus AI addresses the following practical problem dimensions:

- Information fragmentation: course resources, schedules, announcements, discussions, and learning activities are not necessarily presented in one coherent student journey.
- Role and faculty ambiguity: academic information should be scoped to the user's verified university role and faculty; unsuitable authorization may reveal resources or operations to the wrong users.
- Disconnected engagement: social communities, questions, research projects, and course activities often lack a common academic context.
- Limited course-grounded assistance: a generic AI chat response may not refer to the assigned lecture or provide a traceable page-level reference.
- Administrative complexity: verification approvals, content reports, events, and faculty-level management require explicit responsibility and review paths.

The problem affects students who need dependable academic access, professors who manage teaching activities, and administrators responsible for verification and content governance. The project proposes a single role-aware platform with connected academic and social services. Claims such as increased student achievement or reduced administrative workload remain hypotheses until measured in a controlled evaluation.

1.3 Importance of the Project

- Unified digital academic environment: integrate course delivery, social communication, research collaboration, and university services in a common, faculty-aware platform.
- Improved access and educational organization: provide structured routes to lectures, assessments, grades, resources, and academic schedules, reducing avoidable navigation between independent systems.
- Stronger academic interaction: connect students, professors, and administrators through community discussions, project sharing, announcements, consultations, and real-time communication.
- Responsible intelligent learning support: enable course-based AI assistance, practice-question generation, and personalized study guidance while considering source attribution, privacy, and institutional oversight.

1.4 Project Objectives

❖ The main objective of **NexaCampus AI** is to design, develop, and evaluate an intelligent academic community platform that integrates learning management, academic social networking, research collaboration, and artificial intelligence into a unified digital environment.

The project aims to achieve the following measurable and verifiable objectives:

- **Develop a unified academic platform** that integrates educational, social, and administrative services into a single accessible system, verified through successful execution of the main student, professor, and administrator workflows.
- **Establish secure and faculty-specific academic environments** supporting six faculties and four user roles (Student, Professor, Faculty Admin, and System Admin), with account verification, appropriate faculty redirection, and role-based access permissions validated through functional tests.
- **Implement a comprehensive Learning Management System (LMS)** that enables course creation, student enrollment, lecture organization, assignment submission, online examinations, grading, and academic progress monitoring, verified through complete teaching and learning workflows.
- **Enhance academic communication and collaboration** by providing social networking, real-time messaging, academic communities, discussions, project sharing, and research collaboration tools, with each major interaction validated through end-to-end testing.
- **Integrate AI-powered academic assistance** capable of answering course-related questions using uploaded educational materials, generating practice quizzes, and producing personalized study plans, evaluated through response relevance, source-reference accuracy, and functional correctness.
- **Improve academic and administrative management** by supporting targeted announcements, academic events, attendance tracking, consultation

appointments, notifications, and administrative moderation, verified through role-specific operational scenarios.

- **Ensure system security, usability, and reliability** by enforcing authenticated access, protecting academic resources, providing responsive interfaces, and evaluating system performance against defined acceptance criteria, including a target response time of under three seconds for common operations under specified test conditions.
- **Evaluate the overall functionality and effectiveness of the platform** through requirements traceability, integration testing, system testing, and user acceptance scenarios to determine whether the implemented features satisfy the project's functional and non-functional requirements.

1.5 Project Scope

The scope of **NexaCampus AI – Intelligent Academic Community Platform** encompasses the design and development of a unified web-based academic environment that integrates learning management, academic social networking, research collaboration, artificial intelligence, and university administrative services. The platform is designed primarily for the Syrian Private University and aims to support students, professors, faculty administrators, and system administrators through role-based access to academic and social services.
The system covers six university faculties and provides faculty-specific academic portals that organize educational content and services according to each user's academic affiliation. It includes user authentication, university account verification, profile management, course creation and enrollment, lecture and resource management, assignments, online examinations, grading, and academic progress tracking.
Furthermore, the platform supports academic and social interaction through posts, stories, communities, real-time messaging, academic discussions, project sharing, research reference management, and collaborative documents. It also includes academic announcements, event registration, calendars, attendance management, office-hour appointments, notifications, and administrative content moderation.
An important part of the project is the integration of **Nexa AI**, which aims to provide intelligent academic assistance through course-based question answering using Retrieval-Augmented Generation (RAG), AI-generated practice quizzes, personalized study planning, and educational recommendations. These services are intended to support students in understanding course materials and improving their learning experience.

From a technical perspective, the project focuses on a responsive web application developed using **React, Node.js, Express.js, and MongoDB**, following a client–server architecture. It incorporates authentication, role-based authorization, academic data management, and controlled access to educational resources.

Project Boundaries and Limitations

The platform focuses on supporting academic activities and communication rather than replacing the university's official administrative information systems. University identity verification is based on the specified student identification rules and administrative approval, rather than direct integration with official university enrollment databases.

The project does not include financial transactions, tuition payment processing, or direct management of official university records unless such integrations are developed separately. AI-generated responses and educational recommendations are intended to support learning and do not replace professors' academic judgment or official assessment decisions.

The project's primary deliverable is a responsive web-based platform. Native mobile applications, university-wide production deployment, and external institutional system integrations are considered potential future extensions rather than mandatory components of the current scope.

1.6 Project Contributions and Deliverables

- An extensible academic web prototype with a React client, a Node.js/Express API, and a MongoDB-based data model.
- A consistent distinction between student, professor, faculty administration, and system administration responsibilities.
- A faculty-aware verification journey and academic workflows supported by documented functional requirements and acceptance scenarios.
- A combined learning and community design connecting academic materials, assessments, discussions, projects, and student collaboration.
- An AI integration design for course-grounded responses and learning support, accompanied by explicit evaluation and safety requirements.

- Engineering documentation covering methodology, comparative study, implementation scope, risks, a proposed sprint plan, and test evidence requirements.

1.7 Organization of the Thesis

Chapter 1 introduces the research by presenting background, scientific problem, importance of the project, research objectives, project scope, research contributions, and the overall organization of the thesis.

Chapter 2 defines the fundamental technical concepts, reviews comparable educational platforms and recent literature, and identifies the design gap addressed by NexaCampus AI.

Chapter 3 presents the development methodology, general functional and non-functional requirements, detailed Graduation Project II requirements, feasibility, risks, and an illustrative six-sprint project p

Chapter Two - FUNDAMENTAL CONCEPTS AND RELATED WORK

CHAPTER 2 — FUNDAMENTAL CONCEPTS AND RELATED WORK

2.1 Fundamental Concepts

2.1.1 Learning Management Systems (LMS)

An LMS structures course-oriented activity: enrollment, educational resources, assignments, assessment attempts, grading, and teacher–student feedback. A course usually defines access boundaries, learning objects, schedules, and assessment policies. In NexaCampus AI, this model is extended through faculty-specific portals and academic networking, without assuming that a social post is interchangeable with an assessed learning activity.

2.1.2 Academic Social Networking and Communities

An academic social environment supports personal profiles, connections, conversations, posts, shared projects, and communities of interest. The educational value arises when discussions and project exchanges can be linked to academic identities and, where appropriate, course or faculty context. Community membership and course enrollment remain separate concepts: a public academic community may include participants who are not enrolled in a specific course.

2.1.3 Role-Based Access Control (RBAC) and Faculty Scope

RBAC associates permissions with roles such as Student, Professor, Faculty Admin, and System Admin. Effective authorization also requires the system to evaluate ownership and resource scope: a professor should manage only authorized courses, and a faculty administrator should not receive access to unrelated faculty data. These checks must be enforced by the server for every relevant endpoint rather than only hiding interface controls.

2.1.4 Academic Identity Verification

The verification workflow begins after basic authentication. The student selects a faculty, submits a university identifier, and receives validation feedback. If the identifier satisfies the project rules, a pending request is recorded for administrative approval. While waiting, the interface may poll the request status at a fixed interval; approval unlocks the faculty portal, whereas rejection keeps protected access restricted. Correct identifier syntax is a preliminary validation step, not proof of official identity.

2.1.5 Retrieval-Augmented Generation (RAG)

RAG combines retrieval of relevant external information with language-model response generation. For academic materials, the intended workflow is to process approved lecture files, split and index text, retrieve relevant passages for a student question, and compose a cited answer. Li et al. (2025) review educational RAG implementations and identify hallucination, outdated knowledge, retrieval incompleteness, computation cost, and multimodal limitations as continuing challenges [4]. A source citation should therefore be checked for relevance and factual support rather than treated as an accuracy guarantee.

2.1.6 Responsible Generative AI in Education

Academic AI must preserve student agency, protect learning records and confidential files, and make uncertainty visible. The 2023 UNESCO guidance emphasizes human-centered governance and privacy safeguards when applying generative AI in educational settings [5]. NexaCampus AI consequently treats source-based assistance as study support rather than a substitute for official teaching, examination rules, or professor judgment.

2.2 Related Systems and Recent Research

2.2.1 Moodle

Moodle is a widely used LMS with structured course activities and resources. Official Moodle 5.0 documentation identifies assignments, quizzes, and forums among course activities [1]. Its relevance to NexaCampus AI is its strong course-centered organization. The project's different emphasis is a customized faculty-aware university social layer, student project presentation, and a proposed source-referenced AI assistant. This comparison concerns configuration and intended integration, not an assertion that Moodle cannot be extended.

2.2.2 Google Classroom

Google Classroom provides class organization, assignment workflows, grading, and instructor feedback. Official support documentation describes numeric grades, comments, submission status, and grade return operations [2]. NexaCampus AI uses these workflows as a practical benchmark for course assessment but seeks a broader platform that combines academic identity, research/community tools, faculty administration, and AI within one application.

2.2.3 Microsoft Teams for Education

Microsoft Teams for Education combines collaboration and educational workflows, including assignments and grades. Microsoft's official documentation confirms teacher assignment creation, student submissions, rubrics, and progress tracking [3]. This establishes Teams as a relevant academic communication comparator. The difference

targeted by NexaCampus AI is an institution-specific academic model and faculty-scoped governance rather than an unsupported claim that Teams lacks collaboration functionality.

2.2.4 Li et al. (2025): Educational RAG Survey

Li et al. (2025) synthesize 51 educational RAG studies and review indexing, retrieval, generation, learning applications, and unresolved technical and pedagogical constraints [4]. The study informs the Nexa AI design: retrieval relevance, accurate citations, content freshness, privacy, and the evaluation of answer faithfulness must be treated as first-class concerns.

2.2.5 UNESCO (2023): Guidance on Generative AI in Education and Research

UNESCO's guidance discusses governance, privacy, human oversight, and pedagogically appropriate use of generative AI [5]. These concerns reinforce the need to separate AI-generated study assistance from institutional grade records, to respect user permissions when retrieving course files, and to provide a practical review process for misleading outputs.

2.3 Comparison of Related Approaches

Table 2-1 summarizes documented platform strengths and the integration targets of NexaCampus AI.

Table 2.1 Comparison between related studies

No table of figures entries found.**System / reference**	**Documented primary strengths**	**Aspect targeted by NexaCampus AI**
Moodle	Structured LMS courses, assignments, quizzes, forums, and learning resources [1].	Combine LMS with faculty-specific social/research modules and course-grounded AI.
Google Classroom	Assignments, submissions, grade/feedback workflow [2].	Integrate broader campus services, communities, and four-role administration.
Microsoft Teams for Education	Communication, collaborative class teams, assignments and grades [3].	Deliver a purpose-built academic workflow with faculty-scoped data and portals.
Li et al., 2025	Survey of educational RAG implementations and limitations [4].	Apply retrieval and citation checks to authorized course documents.

No table of figures entries found.**System / reference**	**Documented primary strengths**	**Aspect targeted by NexaCampus AI**
UNESCO, 2023	Human-centered, privacy-aware policy guidance for educational generative AI [5].	Set governance and evaluation requirements for Nexa AI features.

2.4 Research Gap

Based on the analysis of existing educational platforms, including Moodle, Google Classroom, and Microsoft Teams for Education, several gaps have been identified regarding the integration and customization of academic services within a unified university environment. **NexaCampus AI** aims to address these gaps as follows:

1. Fragmentation of Academic and Social Services: Existing platforms provide learning management and communication capabilities, but combining these services with university-oriented social networking, student project sharing, and research collaboration may require multiple tools or additional integrations. NexaCampus AI addresses this gap by bringing these activities together within a single academic community platform.
2. Institution-Specific Academic Management: General-purpose educational platforms do not automatically implement the particular faculty structure, university identification rules, and administrative approval workflow required by the Syrian Private University. NexaCampus AI addresses this need through faculty-specific portals, university account verification, course enrollment management, and differentiated permissions for students, professors, faculty administrators, and system administrators.
3. Integrated AI-Powered Academic Support: Although existing platforms offer AI-assisted educational features, integrating course-material retrieval, referenced academic answers, practice quiz generation, and personalized study planning into one institution-specific workflow remains a relevant development challenge. NexaCampus AI aims to address this through its Nexa AI assistant and Retrieval-Augmented Generation (RAG) capabilities.
4. Integration of Research and Learning Activities: Conventional classroom workflows do not necessarily combine course activities with dedicated academic project portfolios, citation management, research communities, and collaborative documents in a single environment. NexaCampus AI aims to connect these services to encourage academic knowledge sharing and interdisciplinary collaboration.
5. Unified Academic Administration and Monitoring: Managing academic announcements, student performance, attendance, events, consultations, and content moderation across separate services may increase administrative complexity. NexaCampus AI proposes a centralized environment for coordinating these activities while maintaining role-based access control.

In conclusion, the main gap addressed by NexaCampus AI is not the absence of

individual educational features in existing systems, but the need for an integrated, intelligent, and institution-specific academic platform that combines learning, communication, collaboration, and administration within one coherent environme

Chapter Three - PROJECT METHODOLOGY AND REQUIREMENTS ANALYSIS

Chapter Three Project Methodology and Requirements Analysis

3.1 Development Methodology (Scrum Methodology)

The development of the **NexaCampus AI** platform followed the **Scrum** methodology, one of the most widely used Agile software development frameworks. Scrum was selected because it supports iterative development, continuous feedback, effective team collaboration, and incremental delivery of software features. This methodology enabled the development team to divide the project into manageable phases, monitor progress regularly, and respond efficiently to changing requirements throughout the development process.

The project was organized into a series of **Sprints**, where each sprint focused on implementing a specific group of features. During every sprint, the development team analyzed the assigned tasks, implemented the required functionalities, performed testing, and reviewed the completed work before proceeding to the next iteration. This iterative approach improved software quality and reduced development risks.

Scrum Roles

The Scrum methodology defines three primary roles that contributed to the successful development of the NexaCampus AI platform.

Product Owner

The Product Owner was responsible for defining the project requirements, prioritizing the product backlog, and ensuring that the developed functionalities satisfied the project objectives and user needs.

Scrum Master

The Scrum Master coordinated the development process, facilitated communication among team members, ensured that Scrum practices were followed correctly, and removed obstacles that could affect project progress.

Development Team

The Development Team was responsible for designing, implementing, testing, and integrating the different components of the NexaCampus AI platform, including the frontend, backend, database, authentication system, and user interface.

Scrum Workflow in NexaCampus AI

The development process followed these iterative stages:

1. Requirement Analysis
2. Product Backlog Preparation
3. Sprint Planning
4. System Design
5. Implementation
6. Testing and Bug Fixing
7. Sprint Review
8. Sprint Retrospective
9. Deployment

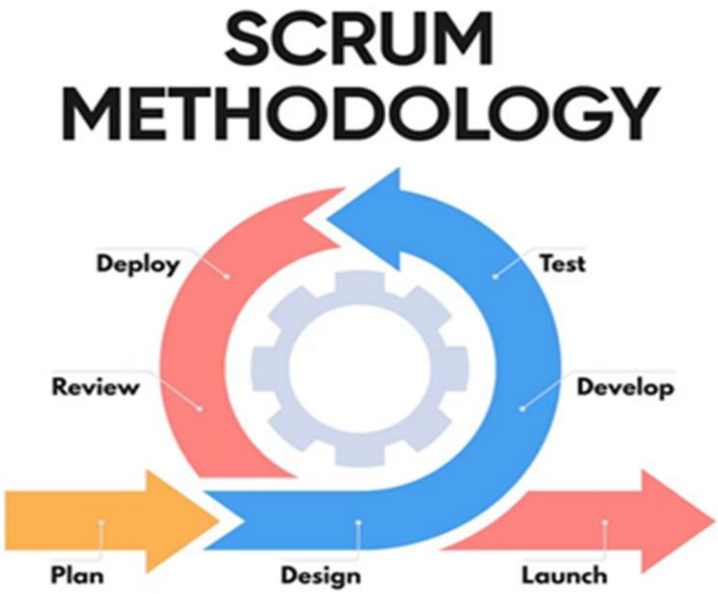


Figure 3.1 Scrum Methodology Process

3.2 General Functional Requirements

General functional requirements define the principal services of the platform. They are intentionally broader than Graduation Project II sprint tasks and cover capabilities inherited from the earlier project stages as well as the new academic extensions.

Table 3.2 Functional Requirments of the NexaCampus AI System

ID	General Function	System Behavior
ID-01	**User Registration**	The system shall allow users to create accounts using their personal information and email addresses.
ID-02	**Login and Logout**	The system shall enable registered users to securely log in and log out of the platform.
ID-03	**Session Management**	The system shall maintain authenticated sessions while users navigate between pages and refresh the application.
ID-04	**Password Recovery**	The system shall support account recovery for users who have forgotten their passwords.
ID-05	**Profile Creation**	The system shall enable users to create profiles containing their names and basic personal information.
ID-06	**Profile Information Update**	The system shall allow users to edit their biography and personal profile information.
ID-07	**Profile Picture Management**	The system shall enable users to upload, replace, and remove their profile pictures.
ID-08	**Profile Customization**	The system shall allow users to personalize profiles with cover images and additional details.
ID-09	**View Member Profiles**	The system shall allow users to access other members' profiles and view information shared publicly.

ID	General Function	System Behavior
ID-10	**Text Post Publishing**	The system shall allow users to create and publish text-based social posts.
ID-11	**Post Image Attachments**	The system shall enable users to attach images to their social posts.
ID-12	**Edit Own Posts**	The system shall allow users to modify the content of posts they previously published.
ID-13	**Delete Own Posts**	The system shall enable users to delete their own posts from the platform.
ID-14	**Social Feed**	The system shall display published posts in a feed that reflects updated social content.
ID-15	**Post Details**	The system shall allow users to open an individual post and view its full content and interactions.
ID-16	**Post Likes**	The system shall allow users to like and unlike posts shared by other members.
ID-17	**Post Comments**	The system shall enable users to add comments to published social posts.
ID-18	**Comment Management**	The system shall allow users to edit and delete comments they have created.
ID-19	**Member Search**	The system shall enable users to search for platform members by name or username.
ID-20	**Profile Post History**	The system shall display a user's published posts and activities on their personal profile.

ID	General Function	System Behavior
ID-21	**Google Single Sign-On**	The system shall allow users to authenticate through Google Single Sign-On (SSO).
ID-22	**Outgoing Connection Requests**	The system shall enable users to send and cancel connection requests to other members.
ID-23	**Incoming Connection Requests**	The system shall enable users to accept or reject received connection requests.
ID-24	**Connection Management**	The system shall allow users to view, organize, and remove established connections.
ID-25	**Following Members**	The system shall enable users to follow and unfollow other members.
ID-26	**Followers and Following Lists**	The system shall allow users to view their followers and following lists.
ID-27	**Social Discovery**	The system shall provide a discovery interface for exploring publicly shared profiles and social content.
ID-28	**Connection Suggestions**	The system shall recommend potential connections based on available social relationships and profile information.
ID-29	**Filtered Member Search**	The system shall allow users to filter member search results by relevant profile attributes.
ID-30	**Post Search**	The system shall enable users to search social posts by keywords and hashtags.
ID-31	**Hashtag Navigation**	The system shall allow users to add hashtags to posts and browse posts associated with a selected hashtag.

ID	General Function	System Behavior
ID-32	**Story Publishing**	The system shall enable users to publish temporary stories containing images or videos.
ID-33	**Story Viewing and Expiration**	The system shall let users view stories and automatically remove expired stories.
ID-34	**Video Posts**	The system shall support publishing video content within social posts.
ID-35	**Multiple Media Attachments**	The system shall allow users to attach and display multiple media items in a single social post.
ID-36	**Saved Posts**	The system shall allow users to bookmark posts and retrieve saved content later.
ID-37	**Post Sharing and Reposting**	The system shall allow users to share or repost publicly available social content.
ID-38	**Report Inappropriate Content**	The system shall enable users to report inappropriate posts or user behavior.
ID-39	**Block and Unblock Members**	The system shall enable users to block or unblock members to prevent unwanted interactions.
ID-40	**Social Privacy Preferences**	The system shall allow users to manage profile visibility and social interaction privacy preferences.

3.3 Non-Functional Requirements

Non-functional requirements specify quality attributes and constraints. Numeric values below are proposed acceptance targets to be validated in a defined test environment; they are not measured outcomes of the current prototype.

Table 3.3 Non-Functional Requirments of the NexaCampus AI System

ID	Quality attribute	Proposed requirement / acceptance direction
NFR-01	Security	The API shall validate authentication and authorization on every protected operation; cross-role and cross-faculty access attempts shall be rejected.
NFR-02	Performance	For a defined local test dataset and workload, at least 95% of ordinary non-AI interactions should complete in under 3 seconds.
NFR-03	Reliability	The system shall return actionable errors for invalid requests, database outages, and network interruptions without corrupting stored records.
NFR-04	Privacy	Personal records, submissions, protected files, and retrieved AI context shall be accessible only to authorized identities.
NFR-05	Usability	Primary student and professor tasks shall be achievable through consistent, responsive interfaces and understandable validation messages.
NFR-06	Accessibility	Critical workflows shall support keyboard navigation, labeled controls, readable contrast, and an audit against WCAG 2.2 criteria.
NFR-07	Scalability	API modules and storage shall be designed to permit future separation; multi-instance SSE requires a shared event backbone.
NFR-08	Maintainability	The project shall separate routes, controllers, models, and services and maintain documented configuration and automated test commands.
NFR-09	File Safety	Uploads shall validate type and access rights; private URLs and malware scanning require verified production configuration.

ID	Quality attribute	Proposed requirement / acceptance direction
NFR-10	Backup and Recovery	Database backup and restoration shall be rehearsed in a test environment before institutional deployment.
NFR-11	Auditability	Administrative moderation, approval decisions, and critical file operations should produce traceable records with timestamps.
NFR-12	AI Answer Quality	Course-grounded responses shall be evaluated for source relevance, citation accuracy, unsupported claims, and permission leakage.

3.4 Functional Requirements for the Current Project

Table 3.4 Current Project Functional Requirments of this NexaCampus AI System

ID	Functional Requirement	Priority
ID-01	Enable students and professors to submit verification requests and allow administrators to approve or reject them	1
ID-02	Control access to system features and academic information according to user roles and permissions.	1
ID-03	Automatically provide users with faculty-specific academic portals and relevant educational content.	1
ID-04	Enable professors to create and manage academic courses, including student enrollment and membership.	1
ID-05	Allow professors to organize course lectures, learning materials, and educational activities within a structured LMS workspace.	2
ID-06	Enable professors to publish assignments, receive student submissions, and provide grades and feedback.	2
ID-07	Enable professors to create, schedule, and evaluate online quizzes and examinations using reusable question banks.	2

ID	Functional Requirement	Priority
ID-08	Allow users to upload, categorize, search, and retrieve academic resources through a centralized knowledge base.	2
ID-09	Enable authorized users to publish and manage announcements targeted to specific courses, faculties, or student groups.	3
ID-10	Allow authorized users to organize academic events and manage participant registration and event capacity.	3
ID-11	Enable users to create, join, and manage academic communities based on shared educational interests.	3
ID-12	Allow students and professors to post academic questions, provide answers, and participate in organized discussions.	3
ID-13	Enable users to exchange private messages and share attachments through real-time conversations.	3
ID-14	Allow students to publish academic projects, showcase achievements, and receive feedback from other users.	4
ID-15	Enable users to collect, organize, and format academic research references for citations and bibliographies.	4
ID-16	Allow authorized users to collaboratively create and edit academic documents while tracking document revisions.	4
ID-17	Enable professors to publish consultation availability and allow students to book available office-hour appointments.	4
ID-18	Enable Nexa AI to answer course-related questions using retrieved academic materials and provide source references.	5
ID-19	Allow students to generate AI-powered practice quizzes from selected academic topics and course materials.	5
ID-20	Generate personalized study plans and recommend relevant learning resources based on student goals and academic progress.	5

ID	Functional Requirement	Priority
ID-21	Enable professors to analyze student performance, assessment results, and learning progress through academic dashboards.	5
ID-22	Allow professors to schedule academic activities, manage class timetables, and record student attendance.	6
ID-23	Automatically notify users about academic announcements, assignments, grades, events, and important system activities.	6
ID-24	Enable administrators to manage user accounts, review reported content, and perform academic content moderation.	6
ID-25	Allow authorized users to securely upload, store, organize, and access academic files according to access permissions.	6

3.6 Project Schedule (Gantt Chart)

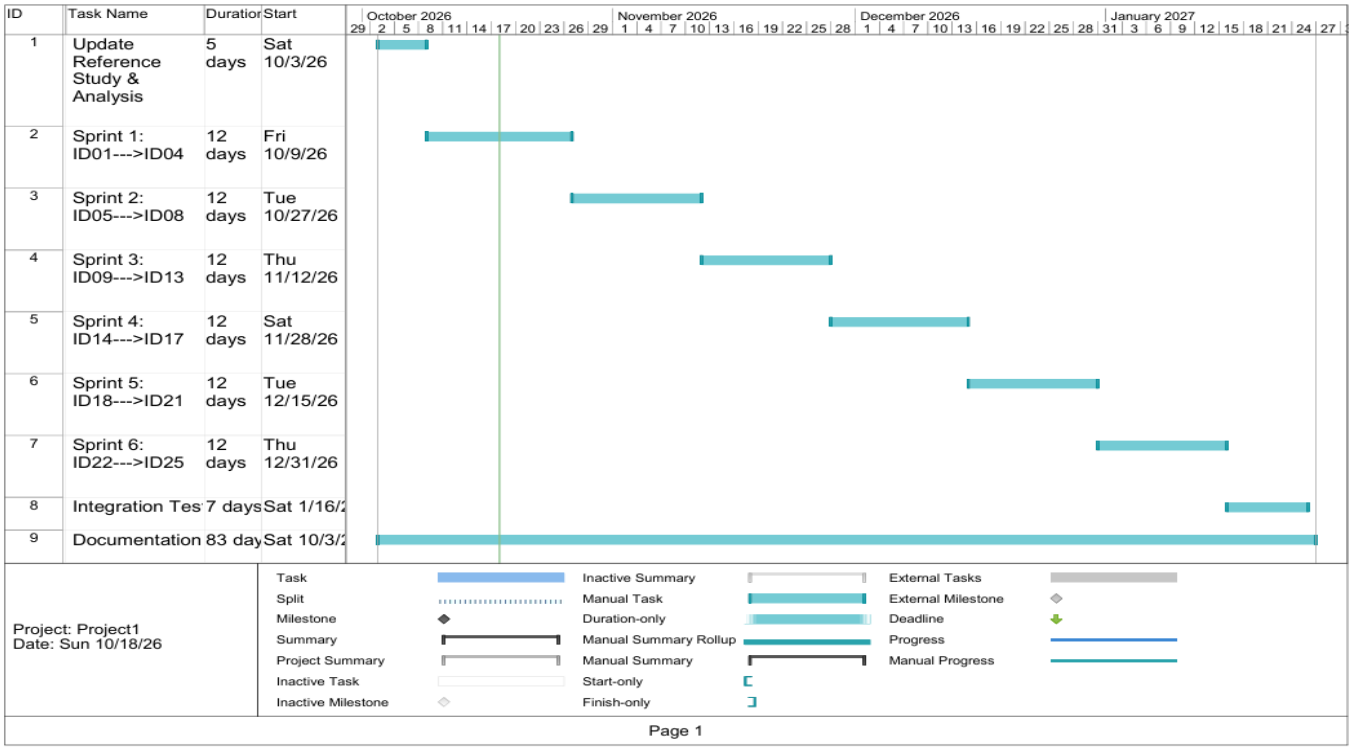


Figure 3.2 Project Schedule (Gantt Chart)